

User Input

Generate a raw QoT dataset for all NDFF paths starting from Bristol using 8 Voyager transmitters only. Use 8 channel slots and enumerate all binary on/off channel patterns across the 8 slots. Simulate only QPSK and 16QAM.



Natural language task

Agentic AI Layer

Policy_update.json/ constraints.json

Planner Agent (LLM)

1. Parsing user input and define research tasks;
2. Generating research scenarios (Scenario Id, simulation case, simulation tool, KPIs).

plan.json

Scenario Expander Agent (LLM)

1. Validate every simulation case against orchestrator contracts;
2. Script Generation.

Results Analysis Agent (LLM)

Validate → aggregate KPIs → compliance → sensitivity → summary

1. KPI history;
2. Simulation logs.

analysis.json

Reflection Agent (LLM)

Gap analysis → attribution → policy updates (search space, constraints, optimizer)

Evaluation results

Report Agent (LLM)

Response user requests. Create detailed report.

report.md / figures

Telemetry / Config / Policy Access

scripts.py / config.json

simulation_results.json

Agent Invocation/ Execution Policies

Execution Status/ Intermediate KPIs/ Agent logs



Network Knowledge Base



Optical Performance Monitoring Data

Network Configuration & Topology Info

DT/Simulation Artifacts & Results

Policies / Constraints / Action Logs

Digital Twin



Physics-based Models

Data-driven ML Models

Decision & Optimization Models

Orchestrator

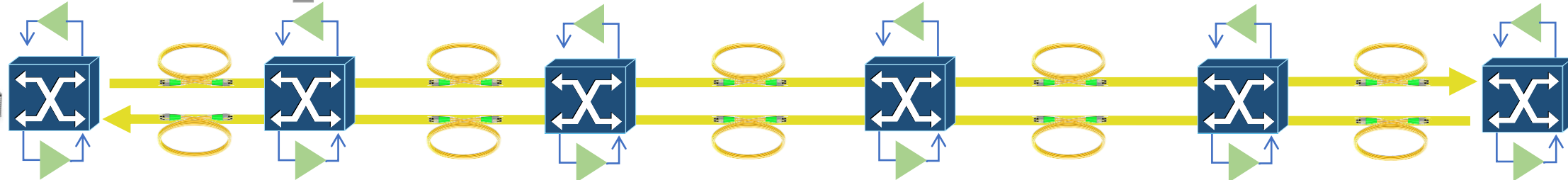
1. Task Graph construction
2. KPI catalogues, Optimisation Strategies
3. Agent coordination
4. State tracking
5. Execution lifecycle management



Telemetry



Transceiver Side



Optical Network Physical Layer



Receiver Side